

Part I. Calculus I competency.

1. (5 pts) Find the equation of the tangent line to the graph of the function $f(x) = \sqrt{7-3x}$ at $x = 1$.
2. (5 pts) Find the derivative of $a(x) = x^{-2/3} + \frac{4}{3x}$
3. (5 pts) Find the derivative of $b(x) = \frac{3x-2}{7-5x}$
4. (5 pts) Find the derivative of $c(x) = e^{-5x} \cos 9x$
5. (5 pts) Find the derivative of $e(x) = x^3 \ln x - 3x^2$
6. (5 pts) Find the derivative of $f(x) = (2x^2 - x + 3)^4$
7. (5 pts) Find the indefinite integral $\int (4x^7 - \sqrt[4]{x^5}) dx$
8. (5 pts) Find the definite integral of $\int_0^{\pi/2} (2x - \cos x) dx$
9. (5 pts) Find the indefinite integral $\int (7-8x)^9 dx$
10. (5 pts) Find the indefinite integral $\int \frac{\sin(\sqrt{x})-1}{\sqrt{x}} dx$

Part II. Calculus II Competency.

11. (5 pts) Find the area the bounded in the first quadrant bounded between the graphs of $y = x^2 + 2x$ and $y = 6x$.
12. (5 pts) Find the indefinite integral $\int x e^{2x} dx$
13. (5 pts) Find the partial fraction decomposition for the function $f(x) = \frac{3x-2}{x^2+4x}$
14. (5 pts) Find the indefinite integral $\int \frac{x-2}{(4x-x^2)^5} dx$
15. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} \frac{\sqrt{k}}{2k^2+1}$
16. (5 pts) Test the series for convergence: $\sum_{k=1}^{\infty} \sqrt{k} \left(\frac{6}{5}\right)^{k-1}$

17. (5 pts) Find the first four non-zero terms of the MacLaurin series of $f(x) = \sin(2x^2)$

Part III. Calculus III competency.

18. (5 pts) Find f_x, f_y for $f(x, y) = xe^{xy^2}$
19. (5 pts) Find f_{xx}, f_{yx} for $f(x, y) = xe^{xy^2}$
20. (5 pts) Find f_x, f_y, f_z for $f(x, y, z) = \frac{y + \sqrt{z - x}}{z^3}$